

Te-Hsun Tsai

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EDUCATION

University of Washington <i>Master of Science in Statistics (Advanced Methods & Data Analysis)</i> (GPA: 3.88/4.0) Ph.D.-level Pass — UW Statistics Theory Exam	Seattle, WA Sep. 2025 – Expected Mar. 2027
National Chengchi University <i>Bachelor of Science in Statistics</i> (GPA: 3.88/4.0)	Taipei, Taiwan Sep. 2021 – Jun. 2025

RESEARCH INTERESTS

Statistical Machine Learning | Biomedical Data Science | Computational Neuroscience | Neural Signal Analysis | Multimodal Representation Learning | Biosignal Representation Learning

RESEARCH EXPERIENCE

Research Intern — Speech–Brain Representation Alignment <i>Academia Sinica, Taipei, Taiwan</i> Advisor: Prof. Yu Tsao	Jun. 2026 – Present
- Developed neural encoding pipelines to investigate how pretrained speech and language models align with human brain responses across EEG and fMRI data. - Evaluated layer-wise representations from WavLM and GPT-2 using ridge regression, leave-run-out cross-validation, noise-ceiling normalization, and robust subject- and channel-level aggregation across clean and noisy speech conditions. - Analyzed complementary EEG and fMRI datasets and explored multimodal strategies for integrating EEG’s high temporal resolution with fMRI’s higher spatial resolution in speech–brain representation modeling. - Extended the pretrained-model analysis toward brain-guided fine-tuning, using neural encoding performance to identify candidate model layers and representations for adaptation.	
Research Assistant — Cause-of-Death Modeling <i>University of Washington</i> Research Supervisor: June Yang, PhD	Sep. 2025 – Dec. 2025
- Developed machine-learning benchmarks (SVM, Random Forest) for automated cause-of-death classification across multiple verbal autopsy datasets (CHAMPS, MDS, PHMRC), establishing reproducible baselines for VA research. - Designed hierarchical classification architectures with soft pruning and chapter-to-detail constraints, enabling stable fine-grained prediction under highly imbalanced, long-tailed medical label distributions. - Optimized model performance through systematic hyperparameter tuning across age-stratified cohorts, balancing VA text representations and structured clinical features via weighted fusion strategies. - Built strict no-leakage evaluation pipelines for heterogeneous data sources (VA narratives, symptom checklists, MITS diagnostics), addressing clinical data sparsity and small-sample challenges.	
Undergraduate Researcher — Forecasting & Data Assimilation for Chaotic Systems <i>National Center for Theoretical Sciences (NCTS), Taiwan</i> Advisor: Prof. Feng-Nan Hwang	Jul. 2024 – Aug. 2024
- Improved forecasting accuracy by integrating Kalman-filter-based data assimilation with LSTM and ARIMA predictors, substantially reducing RMSE across Lorenz-63 experiments. - Conducted 200+ experiments on Lorenz-63 to evaluate the effects of model architecture, training-data size, forecast horizon, and observation noise on closed-loop forecasting performance. - Evaluated Kalman-filter-based data assimilation under noisy observations and model misspecification, demonstrating improved forecast robustness across multiple experimental settings.	

SELECTED PROJECTS

Causal Effect of Private Insurance on Flu Vaccination — Graduate Course Project	
- Estimated the causal effect of private insurance on flu vaccination using 13,612 adults from the 2018 Medical Expenditure Panel Survey. - Applied DAG-based confounder adjustment, outcome regression, Hájek IPW, AIPW, bootstrap confidence intervals, covariate-balance diagnostics, and sensitivity analysis for unmeasured confounding.	
Adaptive Conformal Prediction for Chaotic Time Series — Graduate Course Project	
- Demonstrated substantial coverage degradation of split conformal prediction under closed-loop chaotic forecasting and evaluated rolling-window and EWMA adaptive calibration strategies. - Showed that adaptive calibration recovered near-nominal coverage at long forecast horizons, characterizing the resulting coverage–interval-width trade-off.	

TECHNICAL SKILLS

Programming & Tools: Python, R, SQL, MATLAB, C++, SAS, Stata, LaTeX, Git
Machine Learning & Data Science: Regression, SVM, Random Forest, XGBoost, Cross-Validation, Hyperparameter Tuning, Monte Carlo Simulation.
Causal Inference & Uncertainty Quantification: DAG-based Adjustment, Outcome Regression, IPW, AIPW, Conformal Prediction, Bootstrap Inference
Deep Learning & Representation Models: LSTM, Transformers, Mamba (State-Space Models), WavLM, GPT-2, Pretrained Representation Analysis
Time Series & Neural Data: ARIMA, Kalman Filtering, Data Assimilation, Chaotic-System Forecasting, EEG, fMRI, Neural Encoding Models, Speech–Brain Alignment